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Course = B-Tech CSE

Q1. What is database management system? why is it required?

Ans- A database management system is software that allows users to create, store, manage, retrieve and manipulate data in a structured, efficient way.

DBMS is required because

- Reduces data Redundancy
- Avoids data inconsistency
- Provides data security
- Data Integrity
- Easy data retrieval

Q2. Why is DBMS better than the traditional file system?

Ans :- • Reduces data Redundancy

- Avoids data Inconsistency
- Better Data security
- Backup and Recovery
- Data sharing and concurrency control

Q3- What is data abstraction? Explain its levels?

Ans- Data abstraction in DBMS is the process of hiding complex internal details of how data is stored and maintained and showing only the necessary information to users.

Levels of data abstraction:-

① Physical level (Internal Level):-
- The lowest level of abstraction.
- Describes how data is actually stored in memory.
- Includes details like storage structures, indexing, file organization etc.

② Logical Level (Conceptual Level)
- Describes what data is stored in the database.
- Defines relationships among data.
- Does not show Physical storage details.

③ View Level (External Level)
- The highest level of abstraction.
- Shows only required data to specific users.
- Different users can have different views of the same database.

Q4. What is the 3 schema architecture & why is it important?

Ans- The three schema Architecture is the database design framework that separates the database system into 3 different levels to achieve data abstraction and independence.

The three schema are

(i) External schema (view Level)

- Describes how individual users view the data.
- Different users can have different views.
- Hides the rest of the database.

(ii) conceptual schema (Logical Level)

- Describes the overall structure of the database.
- Includes tables, attributes, relationships and constraints.
- Does not include Physical storage details.

(iii) Internal Schema (Physical Level)

- Describes how data is Physically stored in memory.
- Including indexing, storage structure, file organization etc.

Why is it Important:

- Provides data abstraction
- Ensures data Independence
- Improves security.

Q5. What is data Independence? why is it needed?

Ans: Data independence is the ability to modify the database schema at one level without affecting the schema at the next higher level.

Why is it needed?

- Easier database maintenance
- Flexibility
- Reduced Development cost.
- System stability.

Q6. What is an Entity-Relationship (ER) model?

Ans: The Entity Relationship model is a high level conceptual data model used to design and represent the structure of a database visually. It was introdu-ced by Peter chen in 1976.

The ER model describes data using:

- Entities
- Attributes
- Relationships

It is mainly used during the database design phase.

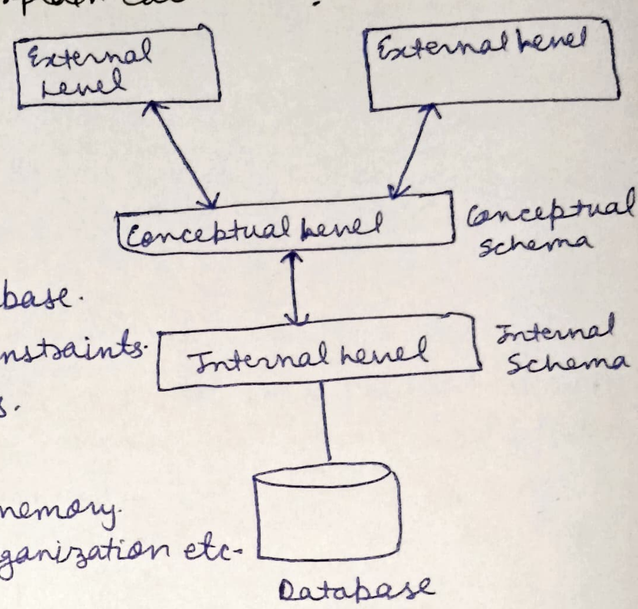
Q6. What are Integrity constraints? why are they important in DBMS?

Ans. Integrity constraints are rules in a DBMS that ensure the accuracy, validi-ty and consistency of data stored in a database. They prevent invalid data from being entered into tables why integrity constraints Important:

- maintain data accuracy
- Ensure consistency
- Improve Reliability
- Prevents data corruption

Section - B

Q1. Draw the three schema architecture and explain each level?



Ans (i) External schema (View Level):

- Describes how individual users view data.
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(ii) Conceptual schema (Logical Level):

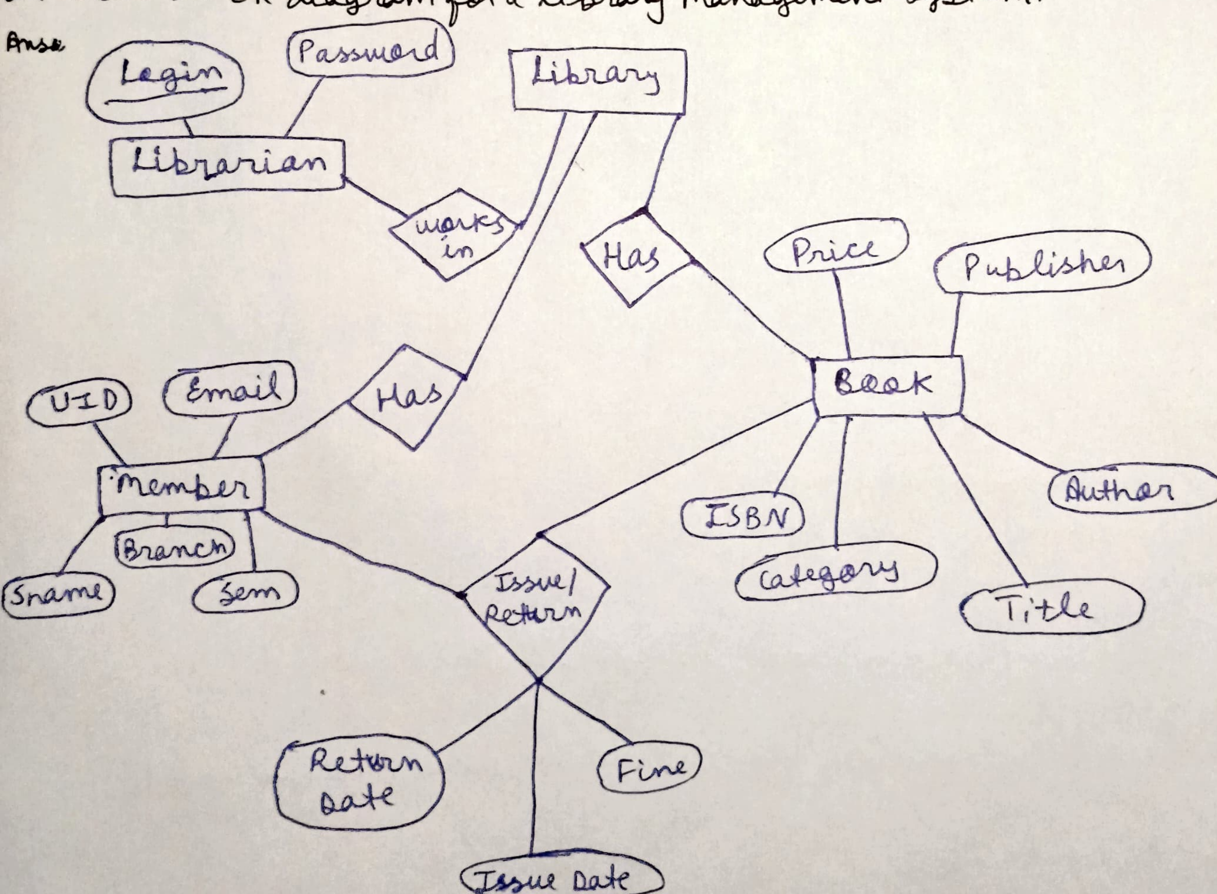
- Describes the overall structure of the database.
- Includes tables, attributes, relationships, constraints.
- Does not include Physical storage details.

(iii) Internal schema (Physical Level):

- Describes how data is Physically stored in memory.
- Includes indexing, storage structure, file organization etc.

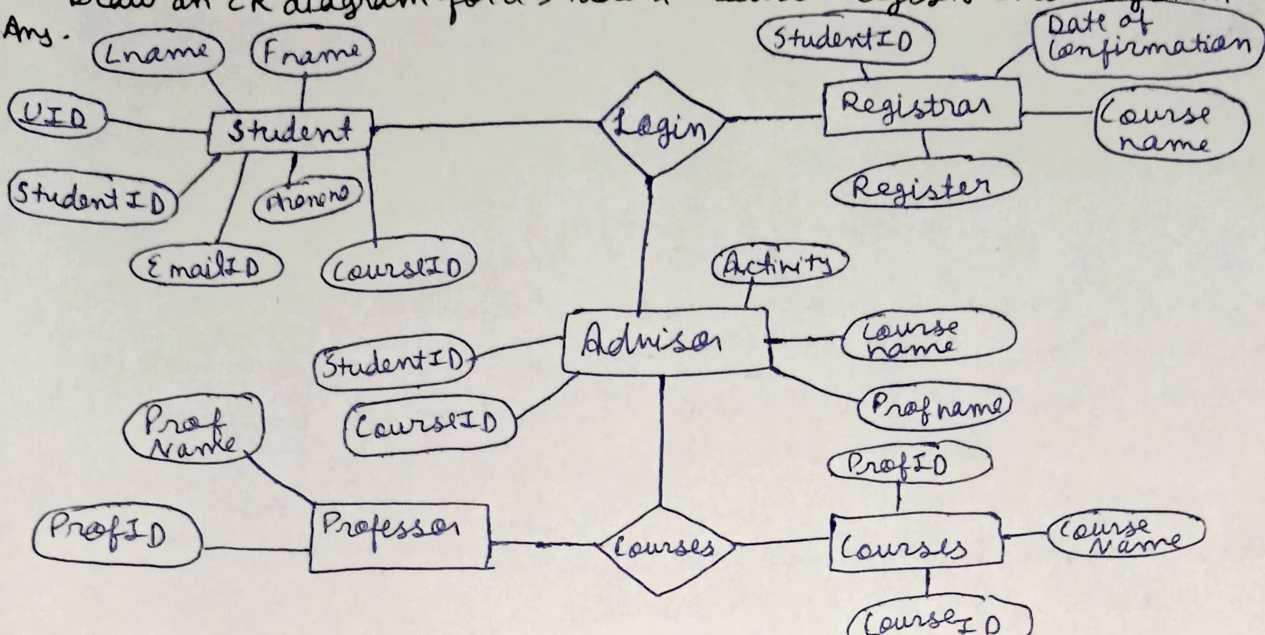
Q2. Draw an ER diagram for a Library management system.

Ans:



Q3. Draw an ER diagram for a Student - Course Registration System.

Ans.



section - C.

Q1. A hospital wants to store patient, doctor, appointment data. Identify entities, attributes and relationships.

Ans ① Entities

(i) Patient
(ii) Doctor
(iii) Appointment

Attributes:-

- Patient_ID
- Name
- Age
- Gender
- Address
- Phone - number

- Doctor_ID
- Name
- specialization
- Phone_number
- experience

- Appointment_ID
- Appointment_Date
- Appointment_Time
- Status

Relationships:

① Patient - Books - Appointment
- A patient can book many appointments
- Each appointment belongs to one patient
- one to many relationship

② Doctor - attends - Appointment
- A doctor can attend many appointments.
- Each appointment is handled by one doctor.
- one to many relationship

Q2. An online shopping platform stores customer, product and order data - suggest keys and constraints.

Ans ① Customer entity:- Customer (customer_ID, Name, Email, Phone, Address)

Keys:-
- Primary key: Customer ID
- Candidate key: Email (should be unique.

Constraints:-
- Customer_ID -> NOT NULL, UNIQUE
- Name -> NOT NULL
- Email -> UNIQUE, NOT NULL
- Phone -> UNIQUE (optional but recommended)

② Product Entity :- Product(Product-ID, Product_Name, Price, stock, category) ③
Keys:-
- Primary Key :- Product_ID

Constraints :-
- Product_ID :- NOT NULL, UNIQUE
- Price :- CHECK (Price>0)
- stock :- CHECK(stock >=0)
- Product_Name :- NOT NULL

③ Order Entity: Order(Order_ID, Order-Date, Customer_ID, Total_Amount, Status)
Keys:
- Primary Key :- Order-ID
- Foreign key: Customer_ID

Constraints:
- Order-ID :- NOT NULL, UNIQUE
- Customer-ID :- FOREIGN KEY
- Total-Amount :- CHECK
- Order-Date :- NOT NULL
- Status :- CHECK

Q3. A university stores students records. Explain how data abstraction helps different users.

Ans (i) View Level (External Level) → For different users : At this level, each user sees only the data relevant to them.

Teacher :-
- Can see : student-ID, Name, course, Marks, Attendance
- cannot see : Fee details.

Accounts Department :-
- can see :- Student_ID, Name, Fee-status
- Cannot see : marks

Student :-
- can see :- own marks, attendance and fee status
- cannot see :- others student's records.

Benefits :- Ensures Security and Privacy.

② Logical Level (Conceptual Level) :-
- Defines the complete structure of the database
- Includes tables like : student (Student_ID, Name, Courses, marks, Fees, Attendance)
- managed by database designers and administrators.

Benefits : maintains overall organization and relationships.

③ Physical Level (Internal Level) :-
- Describes how data is stored in memory.
- Includes Indexing, file organization, storage, blocks, etc.
- Handled by DBMS Internally

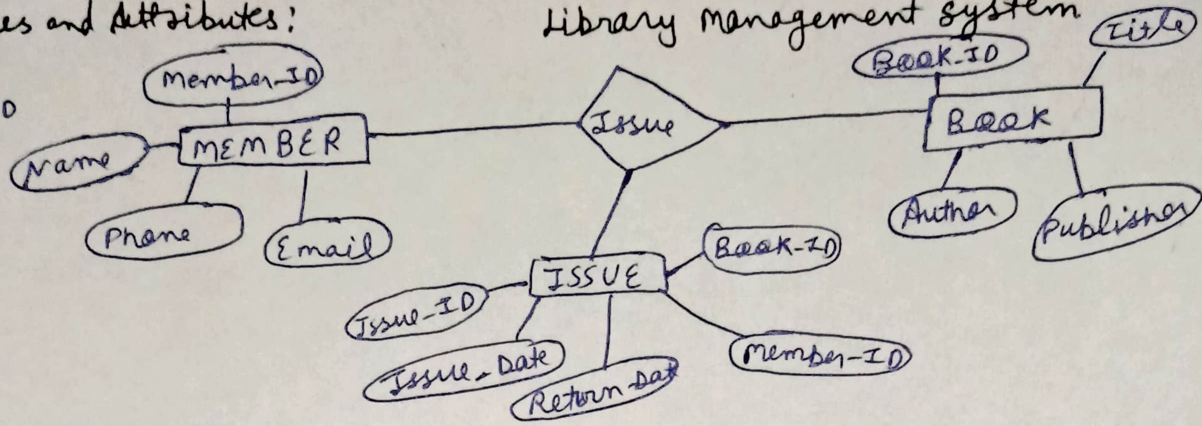
Benefits :- Users do not need to know how data is physically stored.

Section-D.

Q1. Draw a simple ER diagram for any real life system and explain one key design decision.

Ans. Entities and Attributes:

Library Management system



① Member
- Member-ID
- Name
- Phone
- Email

② Book
- Book-ID
- Title
- Auther
- Publisher

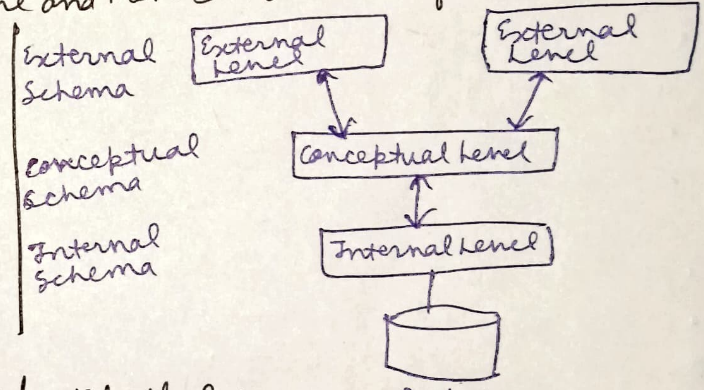
③ Issue
- Issue-ID
- Issue-Date
- Return Date
- member-ID
- Book-ID

Relationships:
- A member can issue many books.
- A Book can be issued by many members (at different times).
- This creates a many-to-many (M:N) relationship b/w member and Book.

To resolve this, we introduce the ISSUE Entity.

Q2. Draw the three schema Architecture and mention one benefit.

Ans. Benefit: Data-Independence: changes made at the physical Level (like changing storage method) do not affect the logical or external levels.

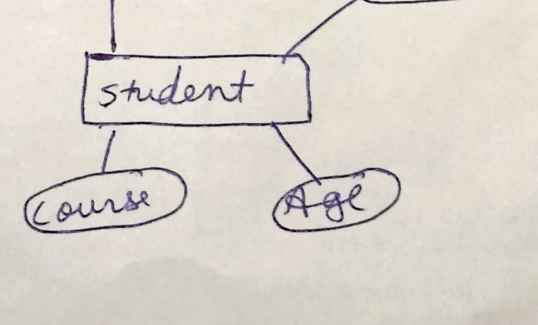


Q3: Draw an Entity with attributes and identify its Primary key;

Ans Explanation: Entity:

Student (represented by a Rectangle)

Attributes:
- Roll-no
- Name
- Course
- Age



Primary Key: Roll-no.
- uniquely identifies each student
- cannot be NULL
- No two students can have the same Roll no.